

Saket Deshmukh

GenAI / AI Engineer

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Summary

GenAI Engineer with 2.5+ years shipping production GenAI applications – RAG pipelines, agentic workflows, and LLM evaluation. Experienced building grounded, guardrailed AI systems using hybrid retrieval, vector search, and self-hosted inference (vLLM), with end-to-end ownership from architecture through Kubernetes deployment and observability.

Skills

GenAI: RAG, Agentic AI, LLM Applications, Prompt Engineering, Guardrails, Grounding

LLMs & LLMOps: LLM Evaluation, LLMops, LangSmith, TruLens, OpenAI API

AI Frameworks: LangChain, LangGraph, FastAPI, REST APIs

Retrieval & Search: FAISS, BM25, pgvector, Hybrid Search, Embeddings, RRF, Semantic Search, Vector Search

Model Serving: vLLM, Self-Hosted Inference, Hugging Face

Programming: Python, SQL, Bash

Databases: PostgreSQL, MySQL

Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, Helm, Terraform, Ansible

CI/CD & DevOps: GitHub Actions, Jenkins, GitLab CI, Git, Linux

Observability: LangSmith, Prometheus, Grafana, AWS CloudWatch

Experience

KWAD Corporation Pvt. Ltd. – AI Engineer

Apr 2025 – Present

AI Compliance & Financial Operations Copilot

Pune, India

- Designed and built an agentic RAG system using **LangChain** and **LangGraph** for natural-language investigation of B2B transaction and compliance data across **12 tenants**, orchestrating multi-step workflows with conditional routing, tool calling, retrieval, response generation, and human-in-the-loop escalation.
- Engineered the RAG pipeline using chunking, embeddings, and **pgvector**, implementing grounded responses with source citations, reducing unsupported/hallucinated answers by an estimated **38%** based on evaluation sampling.
- Implemented tenant-isolated retrieval and input/output guardrails, and moved transactional data access from LLM-generated SQL to parameterized **PostgreSQL/PL-pgSQL** procedures – eliminating a class of SQL-injection and data-leak risk across **12 tenants**.
- Deployed and monitored the system using **LangSmith**, **Docker**, **Kubernetes**, **Helm**, and **GitHub Actions**; tracing and RAG evaluation metrics supported **9 iterations** of prompt/retrieval improvements with **45% faster** mean-time-to-detect regressions.

Briowin Private Limited – Software Engineer

Feb 2024 – Apr 2025

CatalogSage – AI-Powered Product Catalog Assistant

Mumbai, India

- Built an AI-powered **RAG product assistant** enabling natural-language search and recommendations across a **300+ SKU catalog** using **Python**, **FastAPI**, **LangGraph**, and **OpenAI**, reducing average query resolution time by an estimated **52%** vs. manual catalog search.
- Designed a **LangGraph workflow** with conditional routing for input validation, hybrid retrieval, response generation, output validation, and graceful fallback, improving end-to-end request success rate to **94%**.
- Engineered **hybrid retrieval** using **OpenAI embeddings**, **FAISS semantic search**, and **BM25 keyword retrieval**, merging results using **Reciprocal Rank Fusion (RRF)**, improving top-k retrieval accuracy by **27%** over a single-method baseline.
- Built a **RAG evaluation pipeline** using **TruLens** to measure **groundedness**, **context relevance**, and **answer relevance**; containerized the service with **Docker** for deterministic CI testing, catching regressions in **18%** of pre-release builds before ship.

Projects

Self-Hosted LLM Inference Platform

AWS EKS / Kubernetes / Terraform / Docker / vLLM

- Built and deployed a self-hosted LLM inference platform on **AWS EKS**, using **Docker** and **vLLM** for model serving, with **Terraform** provisioning VPC, IAM, and Kubernetes infrastructure.

Education

B.E. in Computer Engineering

Savitribai Phule Pune University

2019 – 2023

Pune, India